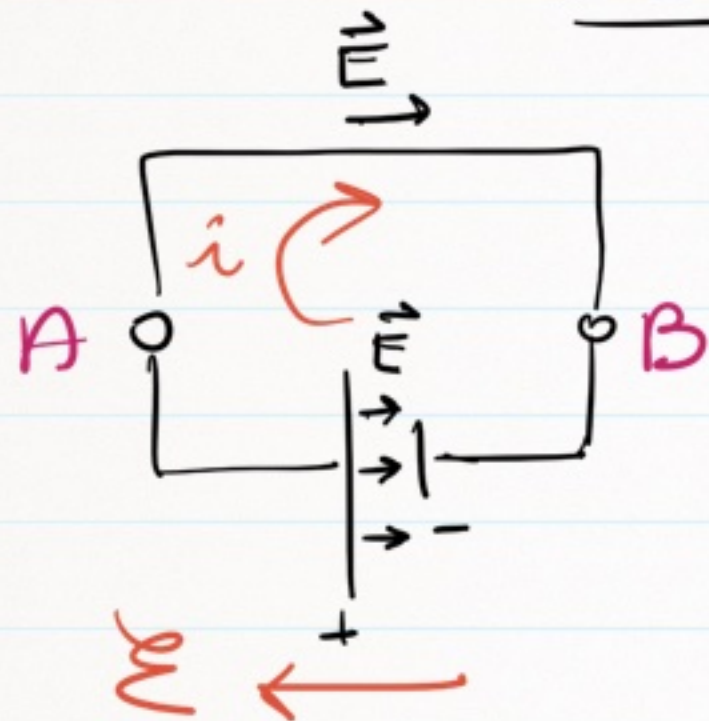


11/11/20

GENERATORE

VAN DE GRAAF (anni '30)
Boston, USA



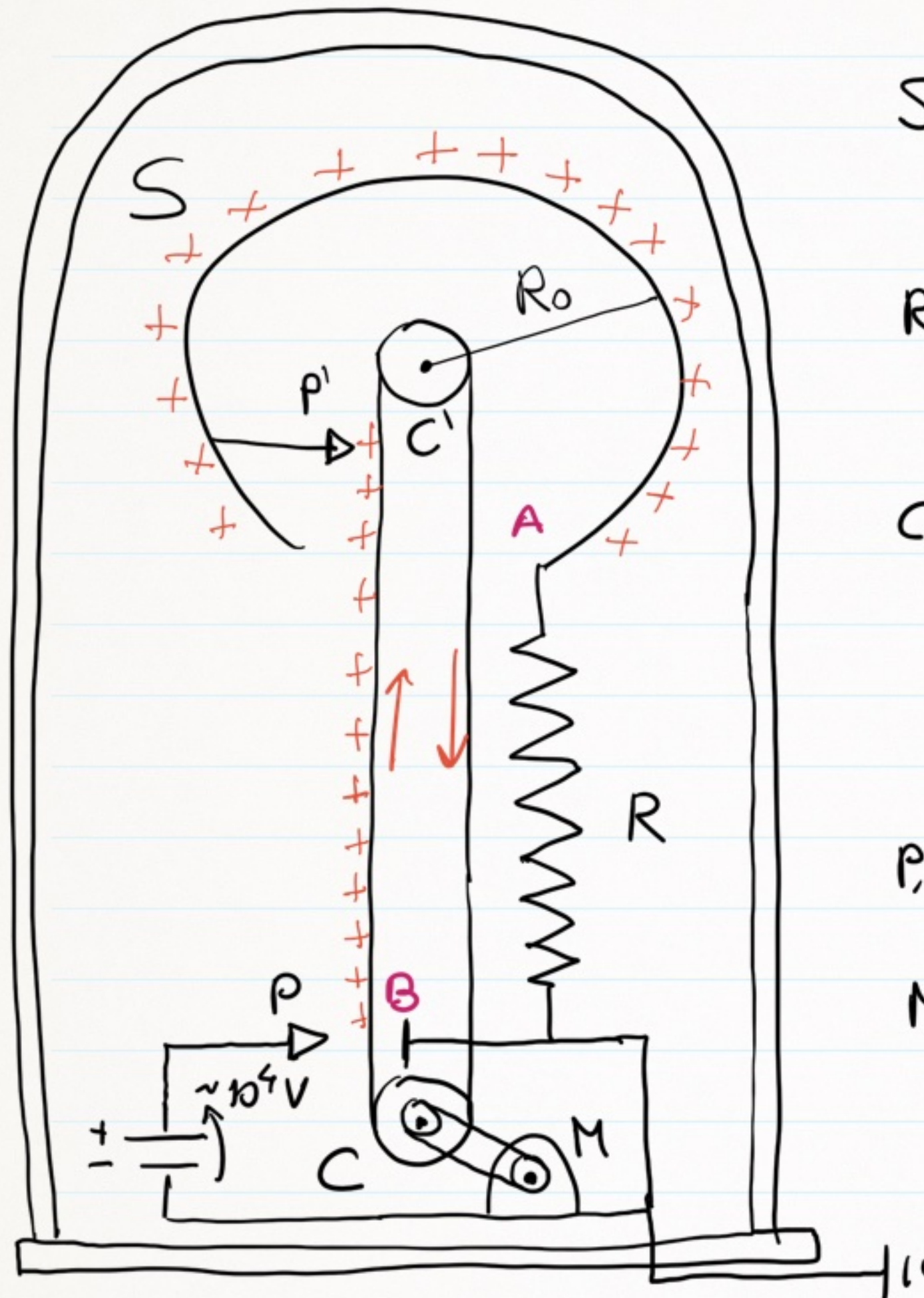
OBBIETTIVO: stabilire una
d.d.p. (continua)
tra 2 elettrodi

COME? spostando cariche
in verso contrario a quello
che sarebbe determinato
dal campo elettrostatico E

→ soluzione MECCANICA

→ sfrutta FENOMENI ELETTROSTATICI {
• induzione
• punte

⇒ Trasformazione di **LAVORO MECCANICO** in
ENERGIA ELETTROSTATICA



S = sfera conduttrice cava
(raggio R_0)

R = resistenza elettrica
della colonna

C, C' = puleggie, tra cui
è tesa una cinghia
(o nastro) isolante
(e.g. caucciù)

P, P' = sistemi di punte

M = motore

con M in funzione, $V_S \rightarrow V_{MAX} = E_{MAX} \cdot R_0$

E_{MAX} ? Rigidità dielettrica del mezzo
circostante (gas nella camera)

e.g. aria $\rightarrow E_S = 3 \cdot 10^6 \text{ V/m}$

se $R_0 = 0.5 \text{ m} \rightarrow E_{MAX} = E_S = V_{MAX} / R_0$
 $\Rightarrow V_{MAX} = 1.5 \cdot 10^6 \text{ V} = \underline{1.5 \text{ MV}}$

con $N_2 @ 10^6 \text{ Pa}$, $V_{MAX} = 6 \text{ MV}$

" $SF_6 @ 7 \cdot 10^5 \text{ Pa}$, $V_{MAX} = \underline{\underline{16 \text{ MV}}}$

massimi valori ottenuti
per una ddp continua

Consideriamo il circuito intorno ai 2 poli **A** e **B** :
q cariche elettriche trasportate

↳ $\vec{F}_m$ = forza meccanica
origiscono

q $\vec{E}$ = " elettrostatica

$$\Rightarrow \vec{F} = \vec{F}_m + q \vec{E} \quad \text{da } \mathbf{B} \text{ a } \mathbf{A}$$

$$\vec{F} = q \vec{E} \quad \text{da } \mathbf{A} \text{ a } \mathbf{B}$$

$$\begin{aligned} W &= \oint \vec{F} \cdot d\vec{S} = \int_B^A (\vec{F}_m + q \vec{E}) d\vec{S} + \int_A^B q \vec{E} d\vec{S} = \\ &= \int_B^A \vec{F}_m d\vec{S} \quad \left[\int_B^A q \vec{E} d\vec{S} + \int_A^B q \vec{E} d\vec{S} = 0 \right] \end{aligned}$$

Forza elettromotrice (f.e.m.) : $\mathcal{E} = \frac{W}{q}$

$q \vec{E}$ aumenta all'aumentare della carica trasportata ed accumulata sulla sfera S

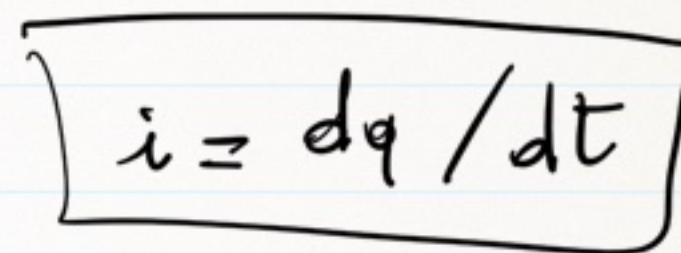
$$q\vec{E}^* = -\vec{F}_m \quad (\text{se } \vec{E}^* < \vec{E}_s)$$

Energia elettrostatica immagazzinata su S :

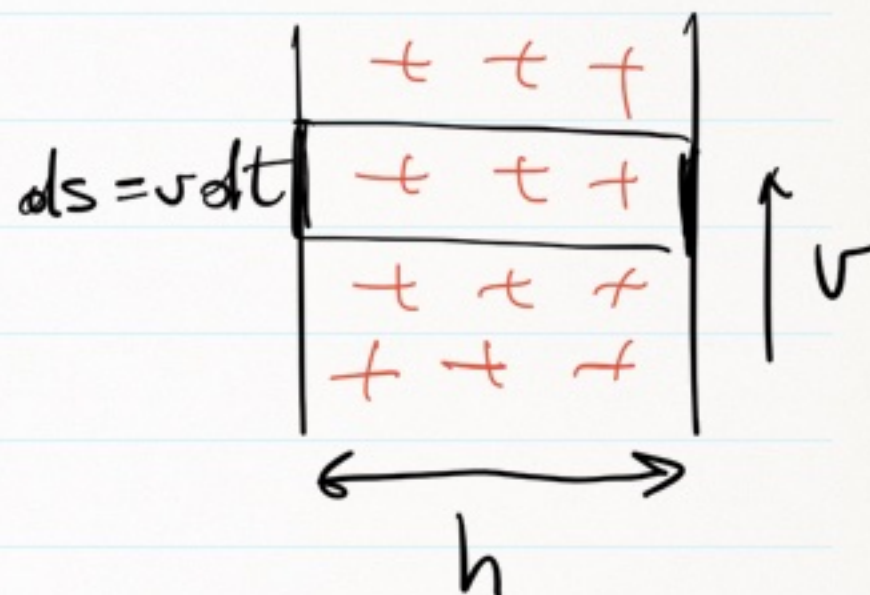
$$E_{el} = \frac{1}{2} C \mathcal{E}^2 = \frac{1}{2} \cdot 4\pi\epsilon_0 R_0^2 \cdot \mathcal{E}^2$$

$$\text{con } \mathcal{E} = 6 \text{ MV (N}_2\text{)}, \quad E_{el} = 2 \cdot 10^3 \text{ J}$$

E la corrente in circolo ?!?



$$i = dq / dt$$



$$i = 100 \mu A$$

$$h = 20 \text{ cm}$$

$$v = 10 \text{ m/s}$$

$$\sigma = 50 \cdot 10^{-6} \text{ C/m}^2$$

$$dq = \sigma d\Sigma = \sigma \cdot h \cdot v \cdot dt$$

$$\underline{i = \sigma h \nu}$$

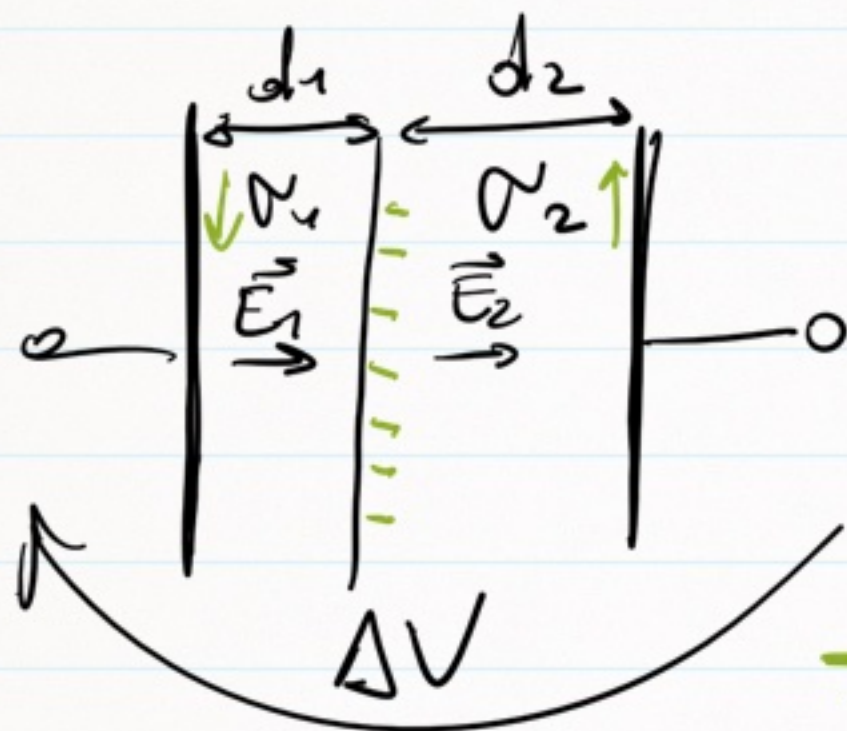
$$R = \mathcal{E}/i = 6\text{MV}/100\mu\text{A} = 6 \cdot 10^{10} \Omega$$

$$P = \sum i = 600 \text{ W}$$

Perché Van de Graaf lo realizzò?!?

e.g. per esperimenti che necessitano di
particelle (ioni) accelerate

Ex. 1

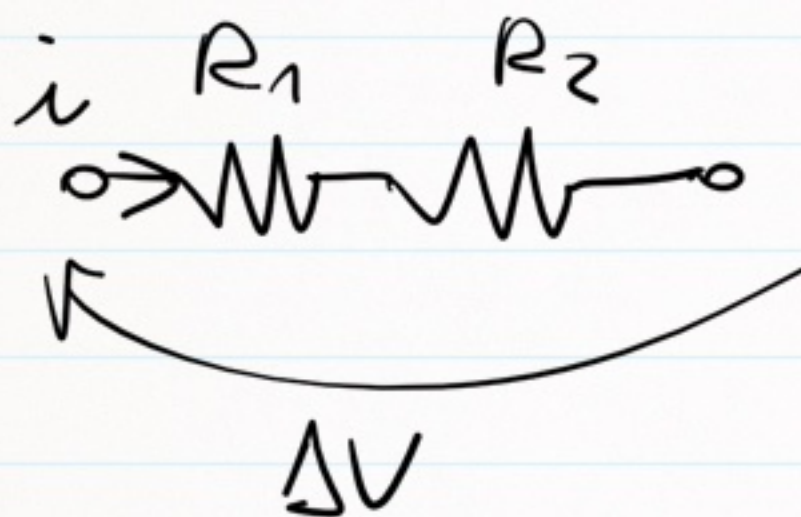


$$\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$$

cond.
stat.

$$\vec{J} = \sigma \vec{E} (= \text{const.})$$

$$\vec{E} = \rho \vec{J} \quad (\rho = \frac{1}{\sigma})$$



$$i = \frac{\Delta V}{(R_1 + R_2)} = \frac{\Delta V}{R_{eq}}$$

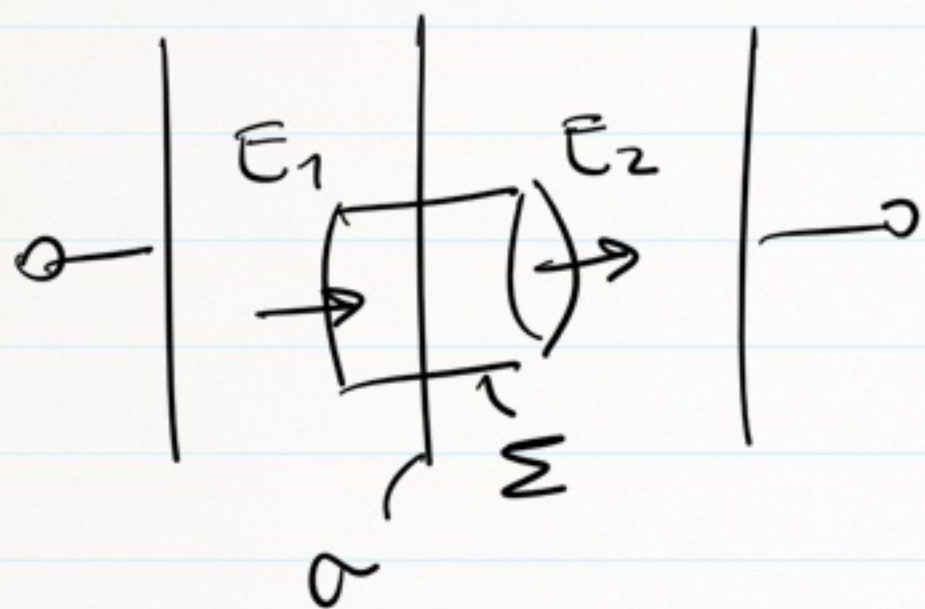
$$R_1 = \rho_1 \cdot \frac{d_1}{A}, \quad R_2 = \rho_2 \cdot \frac{d_2}{A}$$

$$R_{eq} = \frac{d_1}{\sigma_1 A} + \frac{d_2}{\sigma_2 A} = \frac{1}{A} \left(\frac{d_1}{\sigma_1} + \frac{d_2}{\sigma_2} \right) = \frac{1}{A} \left(\frac{d_1 \sigma_2 + d_2 \sigma_1}{\sigma_1 \sigma_2} \right)$$

$$\dot{Q} = \frac{\Delta U}{R_{\text{eq}}} = \frac{\Delta U \cdot A \cdot \sigma_1 \sigma_2}{d_1 \sigma_2 + d_2 \sigma_1}$$

$$J = \frac{\dot{Q}}{A} = \frac{\Delta U \sigma_1 \sigma_2}{d_1 \sigma_2 + d_2 \sigma_1}$$

$$\left\{ \begin{array}{l} \underline{E_1} = \frac{J}{\sigma_1} = \frac{\Delta U \sigma_2 \uparrow}{d_1 \sigma_2 + d_2 \sigma_1} \\ \underline{E_2} = \frac{J}{\sigma_2} = \frac{\Delta U \sigma_1 \downarrow}{d_1 \sigma_2 + d_2 \sigma_1} \end{array} \right.$$



$$\Phi_{\Sigma}(E) = \frac{Q_{int}}{\epsilon_0} = \frac{\sigma A}{\epsilon_0}$$

$$\begin{aligned} \Phi_{\Sigma}(E) &= E_2 A - E_1 A_1 \\ &= (E_2 - E_1) A \end{aligned}$$

$$\Rightarrow \underline{\sigma} = \epsilon_0 (E_2 - E_1)$$

$$= \epsilon_0 \frac{\Delta V}{\sigma_2 d_1 + \sigma_1 d_2} (\sigma_1 - \sigma_2)$$

Ex. 2

$$R_{eq} = R_1 + R_2$$

$$V_1 = R_1 \cdot i = R_1 \cdot \frac{\Delta V}{R_1 + R_2} \\ = 40 \Omega \cdot \frac{60V}{60 \Omega} = 40V$$

$$V_2 = \Delta V - V_1 = 20V$$

$$\rho_1 = R_1 \cdot \frac{A}{L} = 40 \Omega \cdot \frac{0.1 (10^{-2})^2 \text{ m}^2}{40 \text{ m}} = 10^{-5} \Omega \cdot \text{m}$$

$$\rho_2 = R_2 \cdot \frac{A}{L} = 5 \cdot 10^{-6} \Omega \cdot \text{m}$$

$$J = \frac{i}{A} = 10^5 \frac{A}{\text{m}^2}$$

$$E_1 = \rho_1 j = 10^{-5} \Omega \cdot m \cdot 10^5 \frac{A}{m^2} = 1 \frac{V}{m}$$

$$E_2 = \rho_2 j = 0.5 \frac{V}{m}$$

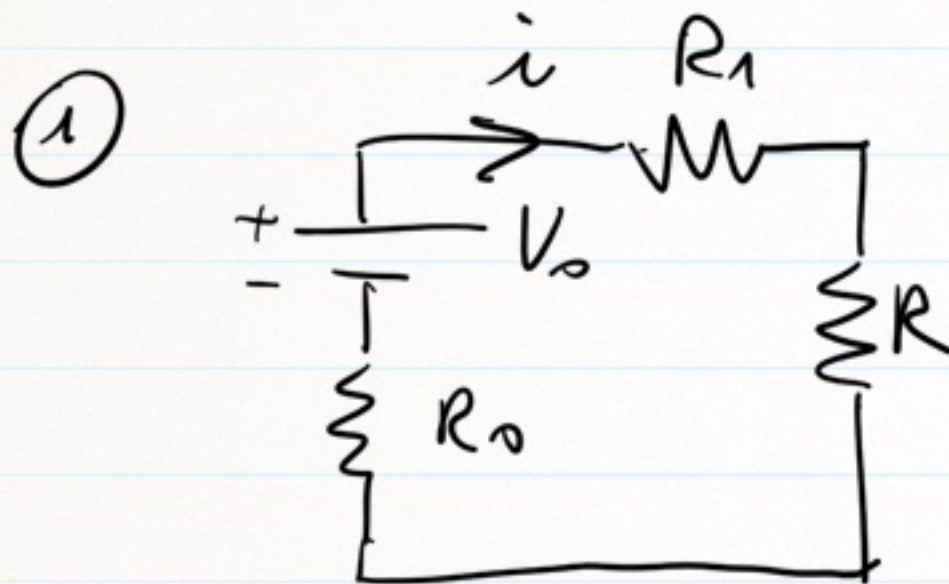
$$q = \sigma \cdot A = (E_2 - E_1) \epsilon_0 \quad A =$$

$$= (0.5 - 1) 8.85 \cdot 10^{-12} \cdot 0.1 (10^{-2})^2$$

$$= - 4.43 \cdot 10^{-17} C$$

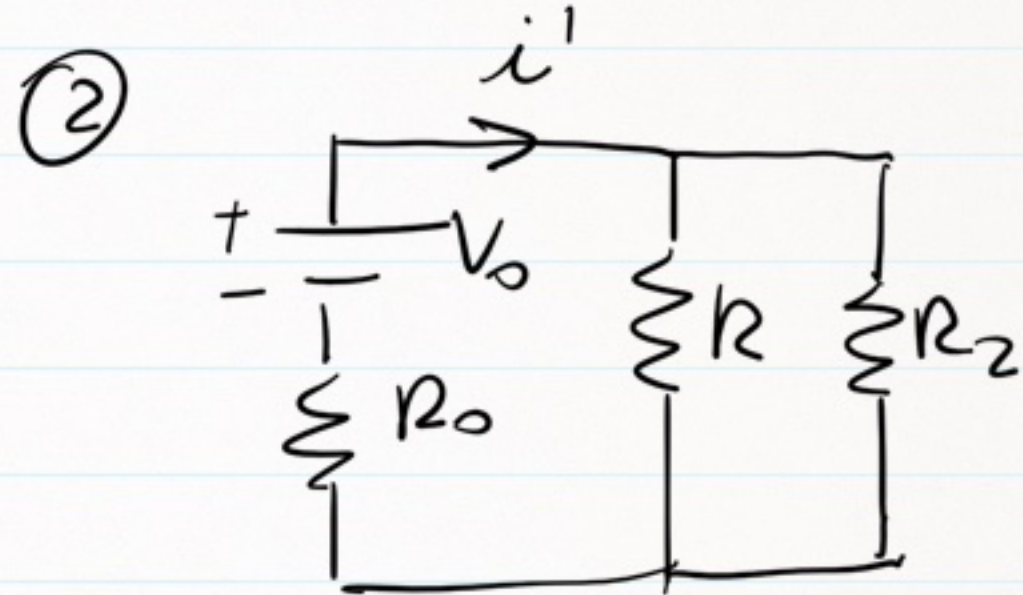


Ex. 4



$$V_0 = R_T \cdot i \rightarrow 1A$$

$$R_T = R_0 + R + R_1$$



$$R_{eq} = \frac{R R_1}{R + R_1}$$

$$V_0 = (R_0 + R_{eq}) i'$$

$$= R_0 i' + \underbrace{R_{eq} \cdot i'}_V$$

$$V = 87V$$

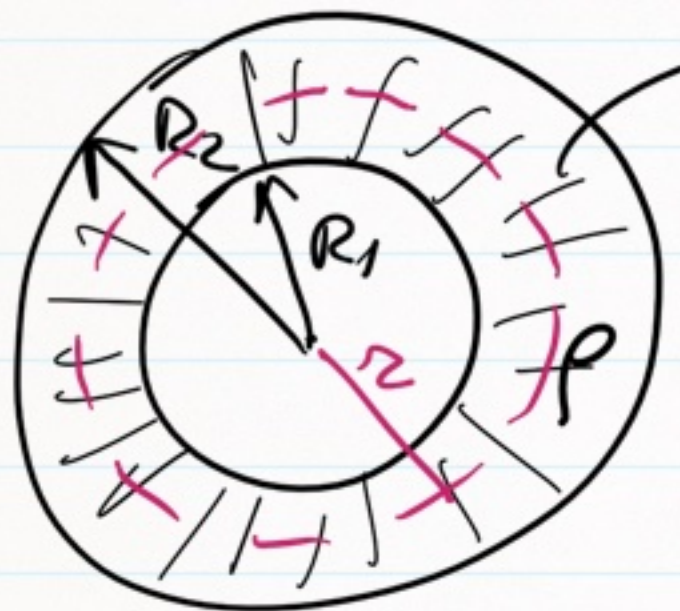
$$i' = \frac{V}{R_{eq}} = 1.26 A$$

$$R_o (\cancel{i'} - i) = \frac{(R_1 + R)i - R_{eq} i'}{(\cancel{i'} - i)}$$

$$R_o = \frac{90 \Omega \cdot 1A - 87V}{(1.26 - 1)A} = 11.47 \Omega$$

$$V_o = (11.47 + 10 + 80) \Omega \cdot 1A = 101.47 V$$

Ex. 5



$R = ?$

$$dR = \rho \cdot \frac{dr}{\Sigma} = \rho \frac{dr}{4\pi r^2}$$

$$R = \int_{R_1}^{R_2} dR = \frac{\rho}{4\pi} \int_{R_1}^{R_2} \frac{dr}{r^2} = \frac{\rho}{4\pi} \left[-\frac{1}{r} \right]_{R_1}^{R_2} =$$
$$= \frac{\rho}{4\pi} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = 25.8 \, \Omega$$

$$i = \frac{V}{R} = \frac{100V}{25.8\Omega} = 3.88A, \quad j = \frac{3.88A}{4\pi(5)^2} = 12.4 \frac{mA}{m^2}$$

Ex. 6 $\vec{E}(r) = \rho \cdot \vec{j}(r)$ LEGGE DI OHM

$$j(r) = \frac{I}{2\pi r h} \Rightarrow E(r) = \begin{cases} \rho_1 \cdot \frac{I}{2\pi r h} & a < r < b \\ \rho_2 \cdot \frac{I}{2\pi r h} & b < r < c \end{cases}$$

$$V_{ac} = \int_a^c E(r) dr$$

$$= \frac{I}{2\pi h} \left(\rho_1 \int_a^b \frac{dr}{r} + \rho_2 \int_b^c \frac{dr}{r} \right)$$

$$= \frac{I}{2\pi h} \left(\rho_1 \ln \frac{b}{a} + \rho_2 \ln \frac{c}{b} \right)$$

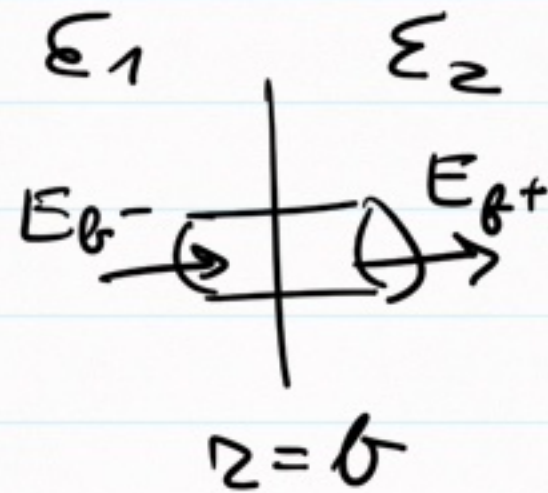
$$I = \frac{2\pi h \cdot V_{ac}}{\epsilon_1 \ln \frac{b}{a} + \epsilon_2 \ln \frac{c}{b}}$$

$$\bar{E}(r) = \begin{cases} \frac{\epsilon_1}{\epsilon_1 \ln \frac{b}{a} + \epsilon_2 \ln \frac{c}{b}} \cdot \frac{V_{ac}}{r} & a < r < b \\ \frac{\epsilon_2}{\epsilon_1 \ln \frac{b}{a} + \epsilon_2 \ln \frac{c}{b}} \cdot \frac{V_{ac}}{r} & \underline{b < r < c} \end{cases}$$

$$(i) @ r = a \quad E_a = \frac{\sigma_a}{\epsilon_1}, \quad \sigma_a = \epsilon_1 E_a$$

$$\underline{Q_a} = 2\pi a h \cdot \sigma_a = 2\pi h V_{ac} \frac{\epsilon_1 \epsilon_1}{\epsilon_1 \ln \frac{b}{a} + \epsilon_2 \ln \frac{c}{b}}$$

$$(ii) @ r = b$$



$$D_{b+} - D_{b-} = \sigma_b, \quad \epsilon_2 E_{b+} - \epsilon_1 E_{b-} = \underline{\sigma_b}$$

$$\sigma_b = \frac{V_{ac}}{b}, \quad \frac{\epsilon_2 \epsilon_2 - \epsilon_1 \epsilon_1}{\epsilon_1 \ln \frac{b}{a} + \epsilon_2 \ln \frac{c}{b}}$$

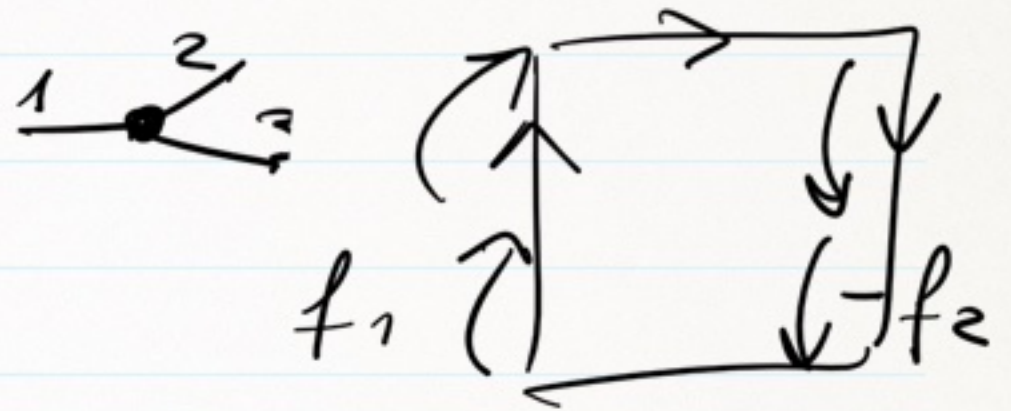
(iii) @ $r = c$

$$\oint_{\Sigma} (\vec{E}) = \frac{Q_a + Q_b + Q_c}{\epsilon_0} = 0$$

$$Q_c = -(Q_a + Q_b)$$

Ex. 7

individuals
- NOD1
- RAM1
- MAGUE



eqs. indep. = # ram1 - # nod + 1

$$\textcircled{1} \quad f_1 - R_1 I_1 - R_2 (I_1 - I_2) - f_2 = 0$$

$$\textcircled{2} \quad f_2 - R_2 (I_2 - I_1) - R I_2 = 0$$

$$I_2 = \dots$$